Independent Project

DC-DC Buck Converter Module

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PCB Details

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Objectives:

Gain an understanding of how a DC-DC Buck Converter Module operates, determine a usable IC chip for the module, and utilize the IC datasheet to create a schematic and PCB design for the module.

Introduction:

This project is a buck converter that was designed around the LM2596 adjustable regulator, capable of stepping down voltages of 4V to 40V down to 1.25V to 35V depending on user configuration. My choice of IC chip is due to my target load of an ESP32 microcontroller because its WiFi/Bluetooth radio draws fast current bursts during transmission, making transient response an important design consideration. This microcontroller was considered specifically because of its importance to my own future personal projects. The goal of this project was to gain an understanding of buck converters and their design as well as create one that I can utilize within my own projects going forward. Furthermore, I also gained hands-on experience with the design process, component selection, and PCB layout.

Details:

The design in Figure 1 follows a typical application circuit recommended in the LM2596-ADJ datasheet, which was selected as a proven, low-risk starting point for a first independent power design project. Given the reliability of the LM2596 platform, the design intentionally builds on the manufacturer's reference application rather than reinventing the topology. The majority of the engineering effort was focused on component-level verification for this specific application and on the adjustable feedback configuration. The LM2596S-ADJ was selected as the regulator with an integrated switch as this simplified the design compared to a discrete controller + external FET approach. The ADJ variant was chosen specifically so the output voltage could be set via an external resister divider. The part is non-synchronous, using an external catch diode rather than a low-side FET which cost some efficiency for simplicity.

C1 provides bulk input capacitance to supply switching current pulses drawn from the input rail and minimize ripple, C2 provides high-frequency decoupling close to the VIN pin to suppress noise the bulk capacitor cannot respond to quickly enough. An SS34 Schottky diode was selected to provide the freewheeling current path for the inductor while the internal switch is off. This diode was chosen for its lower forward voltage drop and faster recovery, both of which reduce switching losses. L1 was selected using the manufacturer's inductor selection guide for this input/output voltage range and target load current. C4 provides bulk output capacitance for ripple reduction and load transient support. C3 provides high-frequency decoupling on the output rail. While RV1 and R3 follow the datasheets standard adjustable-output configuration, the resistor values were calculated specifically for this project using the LM2596's internal reference voltage. The RV1 Potentiometer was utilized in place of a fixed resistor, paired with R3 to allow fine-tuning of the exact output voltage for sensitive microcontroller electronics. An LED was also included (D1) purely as a power indicator.

Moving to PCB Design copper pours were used for the ground plane and high-current power nodes (VIN, VOUT) rather than routing these as narrow traces. This provides a low-resistance path for the DC load current, reduces I^2R losses and heating, and gives the IC's ground pin a thermal path for heat dissipation. The ground pour also provides a low-inductance return path for the switching current, helping minimize noise coupling into the feedback trace and reducing radiated EMI — a relevant consideration given the ESP32's onboard WiFi/Bluetooth radio. The top layer of the PCB can be observed in Figure 2, the bottom layer in Figure 3, and both layers together in Figure 4.

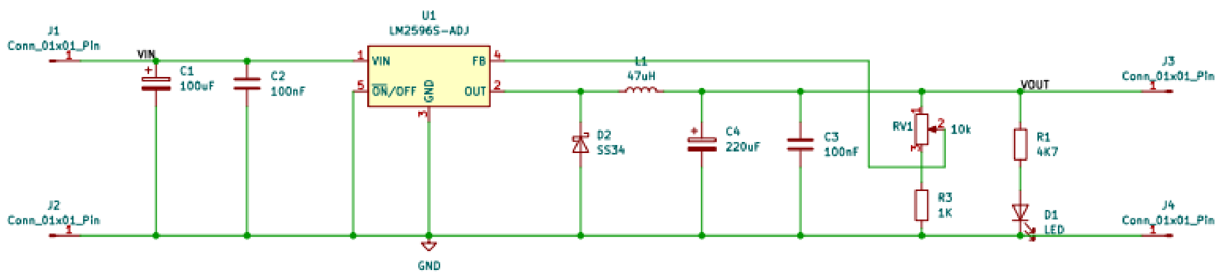

Figure 1: Schematic Design of the Buck Converter

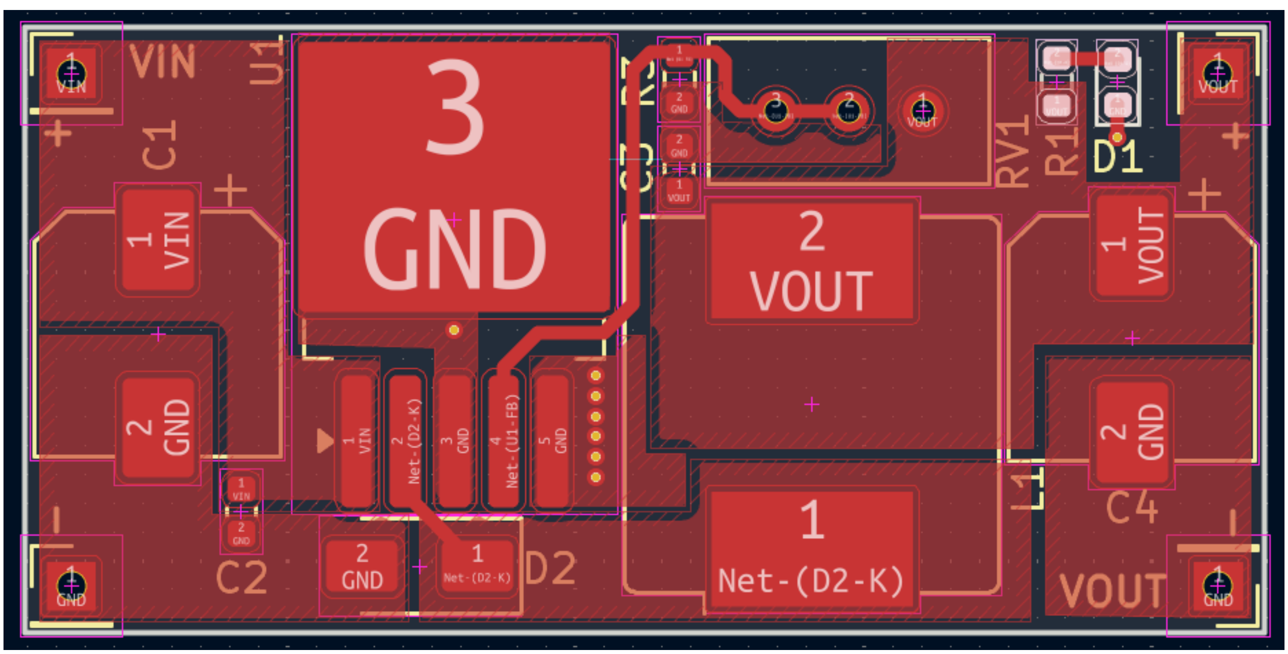

Figure 2: Top Layer Buck Converter PCB Layout

Figure 3: Bottom Layer Buck Converter PCB Layout

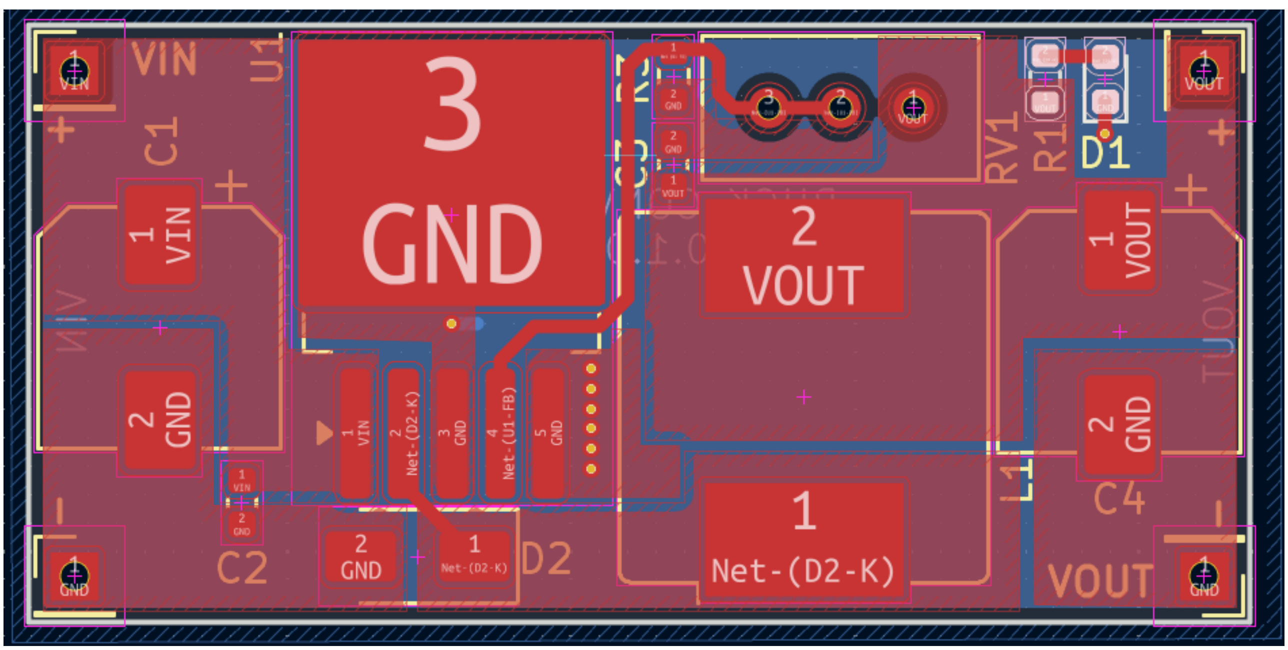

Figure 4: Complete Buck Converter PCB Layout

Conclusion:

This project resulted in a complete schematic and PCB layout for an adjustable buck converter based on the LM2596S-ADJ, targeting a 12V-to-3.3V conversion for ESP32 power delivery. Component selection was based on datasheet guidance and verified against the specific voltage, current, and application requirements of this project, including feedback divider calculations for the target output voltage and layout decisions (copper pours, switch node minimization) aimed at thermal performance and EMI reduction given the ESP32's onboard RF radio.

As this was a design-stage project without physical prototyping, the primary outcomes are the validated schematic design and layout, rather than measured performance data. Key next steps for future work include fabricating and populating the board, and bench testing for output ripple, efficiency, and load transient response under real ESP32 operating conditions, particularly during WiFi transmission bursts, which represent the most demanding load case for this supply.